YEAR 10 PHYSICS, AQA GCSE, VIRTUAL LABS

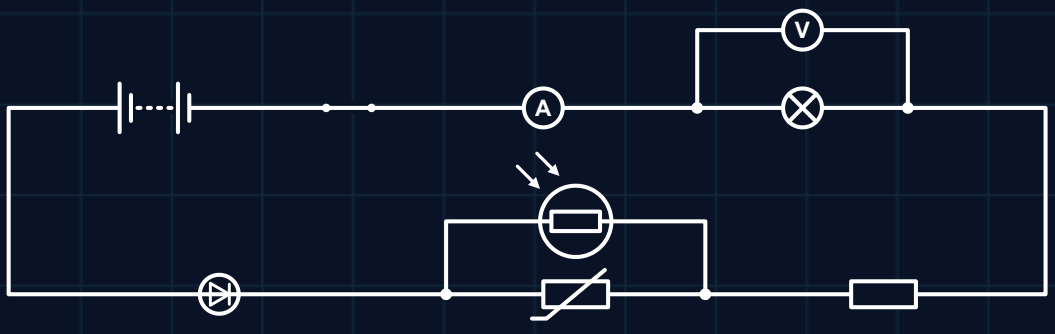
ELECTRIC CIRCUITS

VIRTUAL LABS PREDICT. TEST. EXPLAIN.

Three investigations of charge, energy and resistance, using the PhET Circuit Construction Kit. Work in pairs and swap roles on each new page.

THE THREE LABS

- Lab 1: Is current used up?
- Lab 2: What does a volt mean?
- Lab 3: Discover the resistance rule



NAME _____ PARTNER _____

CLASS _____ TEACHER _____

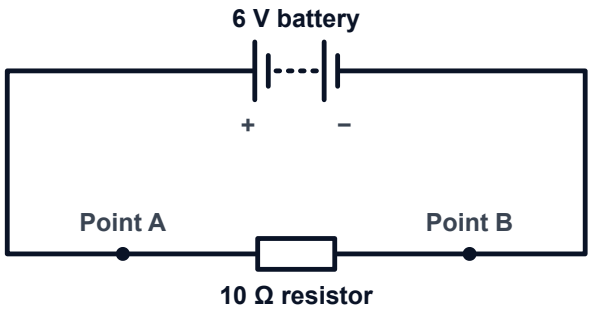
START HERE

SET UP AND METERS

Predict. Test. Explain. Work in pairs. Swap roles on each new page.

> ! Open [PhET Circuit Construction Kit: DC] and choose **Lab**. On paper: search “PhET Circuit Construction Kit DC”, then choose Lab.

1. Select **Lab**. Turn on **Values** and **Show Current**. Start with **Electrons**.
2. Build the loop shown. Join the wire ends together.
3. Tap the battery: set **6 V**. Tap the resistor: set **10 Ω**.
4. In Advanced Options, set Wire Resistivity and Battery Resistance to their **minimum** settings.



One unbroken loop.

MEASURE CURRENT

Drag out the **Non-Contact Ammeter**. Place its probe on a wire. You do not need to break the circuit. A normal wired ammeter goes **in series**.

MEASURE POTENTIAL DIFFERENCE

Drag out a **voltmeter**. Put one probe at each end of the component. Swap the probes if the reading is negative.

KEEP THESE IDEAS SEPARATE

QUANTITY	MEANING	UNIT
Charge Q	An amount of electric charge	coulomb (C)
Current I	Charge passing a point each second	ampere (A)
Potential difference V	Energy transferred per coulomb between two points	volt (V)
Energy E	Energy transferred electrically	joule (J)
Resistance R	Ratio of p.d. to current for a component	ohm (Ω)

> ! The moving dots show flow. One dot is **not** one coulomb. Use the meters. Do not count dots or time their journey.

Lab

01 IS CURRENT USED UP?

Current and charge; Q = I t; workbook Lesson 2

BIG QUESTION **A resistor gets warm when there is a current in it. Does that mean it uses up some of the current?**

PREDICT

PREDICT BEFORE YOU MEASURE

1 Mia makes a claim about the circuit. Do you agree? Explain before measuring.

Mia The resistor uses up some current. There will be less at B than at A.

INVESTIGATE

TEST THE CLAIM

Keep the battery at 6 V and the resistor at 10 Ω. Measure at A, at B and on a third wire near the battery. Move only the ammeter probe.

2 Record the current at each position.

Position	Current I / A
A	
B	
Third position	

3 What do your readings show about current in this single loop? Use numbers in your answer.

BREAK AND RESTORE THE CIRCUIT

Open a connection: select a joint and use its cut control (or Delete). Then reconnect it.

4 What happens to the current when the loop is broken?

EXPLAIN THE FLOW

Switch between **Electrons** and **Conventional**. Look at the wires outside the battery.

5 Complete the sentences.

Electrons move from the _____ terminal towards the _____ terminal.

Conventional current points from _____ towards _____. Does switching the display change the size of the measured current? _____

EXPLAIN IT TO A YOUNGER PUPIL

6 Rewrite Mia's claim using the words **charge** and **energy**.

The charge continues around the circuit. In the resistor, energy …

EXPLAIN

HOW MUCH CHARGE PASSES?

Continue with the same circuit. A current reading describes a rate.

Current is the charge passing a point each second. For a steady current:

Q = I t

charge = current × time

Q in coulombs (C), I in amperes (A), t in seconds (s)

Rearranged: I = Q ÷ t and t = Q ÷ I. One ampere means one coulomb per second.



FROM A METER READING TO AN AMOUNT

> ! **Example:** if I = 0.60 A, then 0.60 C passes a point each second. In 5.0 s, Q = I × t = 0.60 × 5.0 = 3.0 C.

Measure I with a 10 Ω resistor at each battery setting. Calculate the charge passing one point in 10 s and in 20 s. The times are given intervals: no stopwatch is needed.

7 Record the current and calculate the charge.

Battery setting / V	Current / A	Q in 10 s / C	Q in 20 s / C
3			
6			
9			

8 Show one charge calculation, including the unit.

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ANSWER	

THINK ABOUT THE RATE

9 Complete the sentences.

At the same current, doubling the time _____ the charge transferred. For the same time, doubling the current _____ the charge transferred.

10 If you wait for twice as long, does the current itself have to double? Explain.

CHECK

CHECK YOUR UNDERSTANDING

11 A charge of 12 C passes a point in 20 s. Calculate the current.

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12 At 0.30 A, how long does it take for 9.0 C to pass a point?

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13 Sam makes a claim about the reading. Correct it.

Sam 0.60 A means there is 0.60 C in the whole circuit.

Lab

02 WHAT DOES A VOLT MEAN?

Energy per coulomb; E = Q V; workbook Lesson 3

BIG QUESTION **The current is the same all the way round a loop. So what changes when you turn up the battery?**

EXPLAIN

ENERGY PER COULOMB

Potential difference tells us how much energy is transferred **per coulomb**. A p.d. of 6 V across a resistor means 6 J is transferred electrically to the resistor for every 1 C passing through it.

E = Q V

energy transferred = charge × potential difference

E in joules (J), Q in coulombs (C), V in volts (V)

Rearranged: V = E ÷ Q and Q = E ÷ V. One volt means one joule per coulomb.



INVESTIGATE

PREDICT AND MEASURE

1 Predict: for the same 2 C of charge, will more energy be transferred at 3 V or at 9 V? Why?

Use one 10 Ω resistor. At each battery setting, measure the p.d. across the battery, then across the resistor. Touch the two voltmeter probes to opposite ends of the component. Swap the probes if the reading is negative.

2 Record the readings and calculate the energy transferred by 2 C.

Battery setting / V	Battery p.d. / V	Resistor p.d. / V	E for 2 C / J
3			
6			
9			

! Energy is calculated here, not measured by PhET. Use E = V × Q and your **measured** resistor p.d. Small differences from the battery setting are possible.

EXPLAIN POTENTIAL DIFFERENCE

3 Show the calculation for the 6 V setting.

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4 Complete the sentences.

At 9 V, each coulomb transfers _____ J in a resistor with 9 V across it. At 3 V, each coulomb transfers _____ J. The amount of charge in this comparison is the same; the _____ per coulomb changes.

5 A resistor transfers 24 J when 4 C passes through it. Calculate its p.d.

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ANSWER	

A P.D. NEEDS TWO POINTS

6 At 6 V, place both probes on the same piece of wire, then on opposite ends of the resistor.

Readings: same wire _____ V; resistor _____ V.

7 Why is “the voltage *at* the resistor” less precise than “the voltage *across* it”?

FOLLOW THE ENERGY

SHARE THE ENERGY BETWEEN TWO RESISTORS

Set the battery to 6 V. Add a second 10 Ω resistor in series, so there is still only one loop. Label the resistors R_1 and R_2.

8 Predict the p.d. across each resistor, and give a reason.

My prediction: R_1 _____ V; R_2 _____ V.

9 Measure the battery p.d. and the p.d. across each resistor.

Battery p.d. / V	R_1 p.d. / V	R_2 p.d. / V

10 Measure the current on either side of R_1.

Current before R_1 _____ A; current after R_1 _____ A.

ACCOUNT FOR THE ENERGY

Use your voltage readings. Ignore the tiny energy transfer in the wires.

11 Complete the energy account for 2 C of charge.

Where energy is transferred	Calculation E = V × Q	Energy / J
Supplied by the battery		
Transferred to R_1		
Transferred to R_2		

12 Complete the sentences.

The battery supplies about _____ J; the two resistors together receive about _____ J. This supports conservation of _____.

13 Why can the current be the same before and after a resistor even though energy is transferred there?

COMPARE TWO CIRCUITS

Return to one resistor. Measure the current and the resistor p.d. in each circuit below. Use the measured p.d. to find the energy transferred per coulomb.

14 Record your readings.

Circuit settings	Current / A	Measured p.d. / V	Energy per C / J
A: 6 V battery, 10 Ω			
B: 12 V battery, 20 Ω			

15 Compare the charge passing per second and the energy transferred per coulomb in A and B. Explain why current and p.d. describe different things.

Lab

03 DISCOVER THE RESISTANCE RULE

Current, potential difference and resistance; V = I R; workbook Lesson 4

BIG QUESTION **If you know the p.d. across a resistor, can you predict the current before you measure it?**

INVESTIGATE

PREDICT AND COLLECT EVIDENCE

1 Set R = 10 Ω.

Predict: if the p.d. doubles, the current will _____.

Change only the battery setting. Keep the resistor and wires the same. Measure the p.d. across the resistor and the current through the loop at each setting.

2 Record your readings and calculate V ÷ I.

Battery setting / V	Resistor p.d. / V	Current / A	Calculate V ÷ I / Ω
2			
4			
6			
8			
10			
12			

3 Complete the sentences.

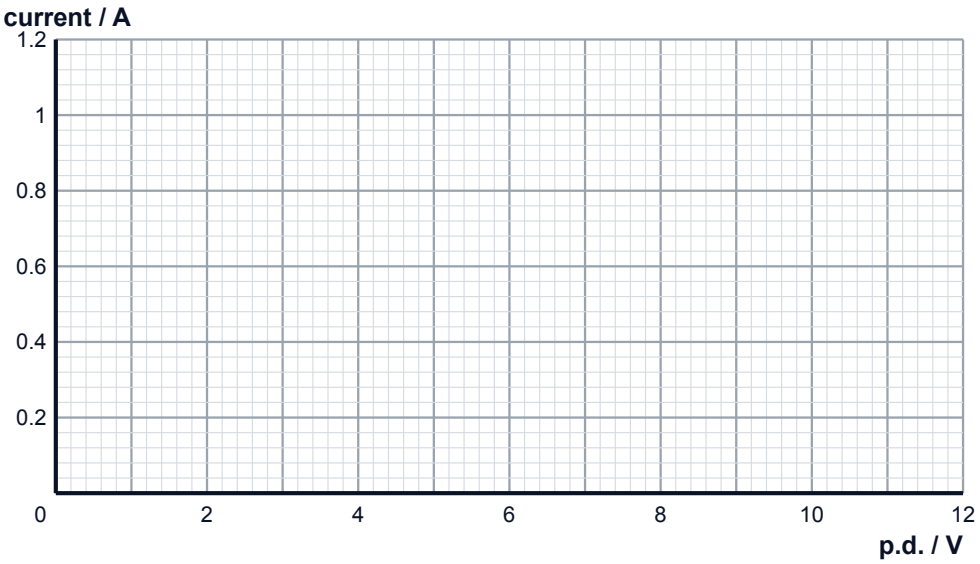
The ratio V ÷ I is approximately _____ Ω. What component setting does it match?

_____

ANALYSE

PLOT YOUR RESULTS

4 Plot p.d. on the horizontal axis and current on the vertical axis. Draw a straight line of best fit if your points support it.



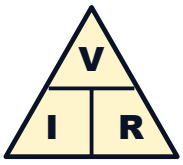
5 Describe the line and whether it passes close to the origin. What does it show about I and V when the resistance stays constant?

V = I R

potential difference = current × resistance

V in volts (V), I in amperes (A), R in ohms (Ω)

For a fixed resistor at constant temperature.



INVESTIGATE

DESIGN A CIRCUIT TO ORDER

Continue Lab 3. Use evidence to choose settings before you test them.

☐ CHANGE THE RESISTANCE

6 Keep the battery at 6 V. Predict before measuring.

If the resistance doubles, the current will _____.

7 Now test these three resistances.

Resistance R / Ω	Resistor p.d. / V	Current / A
5		
10		
20		

8 What pattern do you find? Keep the p.d. approximately constant in your explanation.

! **Rearrange the rule:** I = V ÷ R and R = V ÷ I. Example: at 6 V with 20 Ω, I = 6 ÷ 20 = 0.30 A.

THE DESIGN CHALLENGE

Design a single-resistor circuit that transfers **18 J** to the resistor while **3.0 C** passes through it in **5.0 s**. Work out the settings before building.

9 Work out each quantity.

Find	Equation and working	Answer with unit
Current I		
Resistor p.d. V		
Resistance R		

10 Build your design. Choose a battery setting close to your required resistor p.d.

Measured current _____ A; measured resistor p.d. _____ V. Use those readings to check: Q in 5.0 s = _____ C; E = _____ J.

11 Did your circuit meet the target approximately? Explain any small difference.

CHECK

EXIT TICKET

12 What does a current of 0.40 A mean?

13 What does a p.d. of 8 V across a resistor mean?

14 A 15 Ω resistor has 0.20 A through it. Calculate the p.d. across it.

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15 Which is transferred to the surroundings at a resistor: charge or energy?

CHECK YOURSELF

ANSWERS

Finish each lab first. Then compare your measurements, calculations and explanations with these answers. Meter readings are approximate: small differences are fine if you used your own readings consistently and gave the correct units.

LAB 1: IS CURRENT USED UP?

1. A prediction, so any reasoned answer. The measurements show Mia is wrong: see question 3.
2. About 0.60 A at A, at B and at the third position.
3. The current is the same at every position in the loop, about 0.60 A. It is not used up, so Mia is wrong.
4. The current falls to 0 A. Reconnecting the loop restores it to about 0.60 A.
5. Electrons: negative to positive. Conventional current: positive to negative. No, the measured current stays the same.
6. The charge continues around the loop. In the resistor, energy is transferred to the surroundings. The resistor does not use up charge or current.
7. 3 V: 0.30 A, 3.0 C, 6.0 C. 6 V: 0.60 A, 6.0 C, 12 C. 9 V: 0.90 A, 9.0 C, 18 C. A higher battery setting gives a larger current.
8. For example, 0.60 A for 20 s: $Q = I \times t = 0.60 \times 20 = 12$ C.
9. Doubles; doubles.
10. No. Current is the charge passing each second. Waiting longer lets more charge pass, but the current stays the same.
11. $I = Q \div t = 12 \div 20 = 0.60$ A.
12. $t = Q \div I = 9.0 \div 0.30 = 30$ s.
13. 0.60 A means 0.60 C of charge passes a point each second. It is a rate, not the amount of charge in the circuit.

LAB 2: WHAT DOES A VOLT MEAN?

1. At 9 V. Each coulomb transfers more energy (9 J instead of 3 J), so the same 2 C transfers more.
2. 3 V: about 3 V, about 3 V, 6 J. 6 V: about 6 V, about 6 V, 12 J. 9 V: about 9 V, about 9 V, 18 J. With one resistor, its p.d. is about equal to the battery p.d.
3. $E = V \times Q = 6 \times 2 = 12$ J.
4. 9; 3; energy transferred.
5. $V = E \div Q = 24 \div 4 = 6$ V.
6. Same wire: about 0 V. Across the resistor: about 6 V.
7. A p.d. is a difference between two points, so the voltmeter needs a probe at each end of the component. “At” names one point; “across” names both ends.
8. About 3 V across each. The resistors are equal, so they share the battery's 6 V equally.
9. Battery about 6 V; R_1 about 3 V; R_2 about 3 V. The two resistor p.d.s add up to the battery p.d.
10. About 0.30 A before and after: the current is the same on both sides.
11. Battery: $6 \times 2 = 12$ J. R_1: $3 \times 2 = 6$ J. R_2: $3 \times 2 = 6$ J.
12. 12; 12; energy.
13. Charge is conserved: the charge leaving the resistor each second equals the charge entering it. The moving charge transfers energy in the resistor; the charge itself is not used up.
14. A: about 0.60 A, about 6 V, about 6 J per coulomb. B: about 0.60 A, about 12 V, about 12 J per coulomb.
15. The current is the same (about 0.60 A), so the same charge passes each second. B transfers about 12 J per coulomb, twice as much as A. Current is charge per second; p.d. is energy per coulomb.

LAB 3: DISCOVER THE RESISTANCE RULE

1. Double.

2. 2 V: 0.20 A. 4 V: 0.40 A. 6 V: 0.60 A. 8 V: 0.80 A. 10 V: 1.00 A. 12 V: 1.20 A. The resistor p.d. is about equal to the battery setting, and V ÷ I is 10 Ω every time.

3. 10 Ω. It matches the resistance set on the resistor.

4. The points lie on a straight line from the origin to (12 V, 1.2 A).

5. A straight line through (or close to) the origin. Current is directly proportional to p.d. when the resistance stays constant: doubling one doubles the other.

6. Halve.

7. 5 Ω: about 6 V, 1.20 A. 10 Ω: about 6 V, 0.60 A. 20 Ω: about 6 V, 0.30 A.

8. At the same p.d., a larger resistance gives a smaller current. Doubling the resistance halves the current.

9. I = Q ÷ t = 3.0 ÷ 5.0 = 0.60 A. V = E ÷ Q = 18 ÷ 3.0 = 6.0 V. R = V ÷ I = 6.0 ÷ 0.60 = 10 Ω. Build it with a 6 V battery and a 10 Ω resistor.

10. About 0.60 A and about 6 V. Q = I × t = 0.60 × 5.0 = 3.0 C. E = V × Q = 6.0 × 3.0 = 18 J.

11. Yes. Small differences come from rounding, or from wire and battery resistance that is not exactly zero.

12. 0.40 C of charge passes a point each second.

13. 8 J of energy is transferred for each coulomb that passes through the resistor.

14. V = I R = 0.20 × 15 = 3.0 V.

15. Energy.